

Title

Prevention of Heel Tissue Breakdown in High-Risk Patients: Protocol for a Multi-Site Implementation Trial

Abstract

Rationale: Pressure-related tissue damage at the heel remains a significant complication in acute care settings, yet the optimal timing and approach for prevention across different care phases has not been systematically studied. **Protocol:** This multi-site randomized trial will evaluate whether prophylactic use of a specialized heel protection device reduces the incidence of heel tissue injury in 180 hospitalized patients with neurological conditions or significantly limited mobility. Participants will be enrolled upon arrival at participating ambulance services and followed continuously through emergency departments and hospital ward stays at two academic medical centers in Sweden. Those assigned to the intervention group will receive the protective device applied in the pre-hospital setting and maintained throughout hospitalization; the control group will receive standard positioning and repositioning protocols. Primary outcomes will assess tissue integrity at multiple time-points across the care pathway. Secondary aims will evaluate the feasibility of early implementation in prehospital settings and identify clinical or demographic predictors of tissue breakdown. Risk stratification using validated assessment tools will be performed at each care transition.